Sequences and Series

100 Level Mathematics: every concept explained step by step, with worked examples, jot-worthy notes and practice questions

HOW TO USE THIS GUIDE (so you stop copying whole slides)

1. Read the **Plain English** boxes first. That is the understanding.

2. Copy **only** the yellow **JOT THIS DOWN** boxes into your notebook. Together they are your revision sheet.

3. Do **Test Yourself** at the end without looking, then check the answers.

4. Red boxes are the mistakes lecturers love to set questions on.

What is inside

1	Sequences: limits, ε–N definition, monotone convergence, recursive sequences
2	Progressions (AP, GP, HP), means, sigma notation, standard sums, telescoping, induction
3	Infinite series: partial sums, geometric series, nth-term test, p-series
4	Convergence tests: integral, comparison, limit comparison, ratio, root, alternating, absolute/conditional
5	Power series: radius and interval of convergence
6	Taylor and Maclaurin series, error estimates, binomial theorem and series
7	Common mistakes
8	Test yourself with full answers
9	One-page formula sheet

PART 1 | Sequences

1.1 What is a sequence?

THE IDEA IN PLAIN ENGLISH

A **sequence** is just a **list of numbers in a definite order**: a first term, a second term, a third term, and so on, with no end. Think of a queue at a bank: position 1, position 2, position 3... and each position has one number standing in it.

Formally, a sequence is a **function whose inputs are the positive integers** 1, 2, 3, ... and whose outputs are real numbers. We write it as $\{a_n\}$ or $a_1, a_2, a_3, \ldots$, and a_n is called the **nth term** or **general term**.

Sequence	General term a_n	First few terms
Counting numbers	n	1, 2, 3, 4, ...

Sequence	General term a_n	First few terms
Reciprocals	$1/n$	1, 1/2, 1/3, 1/4, …
Alternating	$(-1)^n$	−1, 1, −1, 1, …
Squares	n^2	1, 4, 9, 16, …
Ratio sequence	$n/(n+1)$	1/2, 2/3, 3/4, 4/5, …
Powers of 2 (plus 1)	$2^n + 1$	3, 5, 9, 17, …

Two ways to define a sequence

- **Explicit (closed) formula:** $a_n = 3n - 2$. You can jump to any term directly ($a_{100} = 298$).
- **Recursive formula:** give the first term(s) and a rule linking each term to the previous ones, e.g. $a_1 = 1$, $a_{n+1} = 2a_n + 1$ gives 1, 3, 7, 15, … You must build up term by term. The **Fibonacci sequence** is $F_1 = F_2 = 1$, $F_{n+2} = F_{n+1} + F_n$.

WORKED EXAMPLE

Finding the general term (pattern hunting). Ask: is it adding the same amount (n multiplied by something)? multiplying by the same factor (a power)? alternating sign (a factor $(-1)^n$ or $(-1)^{n+1}$)?

• 3, 6, 9, 12, … → **$3n$**. • 1/2, 1/4, 1/8, … → **$1/2^n$**.

• 2, −4, 8, −16, … → **$(-1)^{n+1}\,2^n$**. • 2/3, 3/5, 4/7, … → **$(n+1)/(2n+1)$**.

1.2 Monotonic and bounded sequences

Word	Meaning	How to test
Increasing	$a_{n+1} \geq a_n$ for every n	$a_{n+1} - a_n \geq 0$; or $a_{n+1}/a_n \geq 1$ (positive terms); or $f'(x) \geq 0$ where $a_n = f(n)$
Decreasing	$a_{n+1} \leq a_n$ for every n	$a_{n+1} - a_n \leq 0$; or ratio ≤ 1; or $f'(x) \leq 0$
Monotonic	Either increasing throughout or decreasing throughout	One of the above
Bounded above	Some number M with $a_n \leq M$ for all n	Find an upper limit
Bounded below	Some number m with $a_n \geq m$ for all n	Find a lower limit
Bounded	Both bounded above and below ($\lvert a_n \rvert \leq K$)	Both of the above

WORKED EXAMPLE

Show $a_n = n/(n+1)$ is increasing and bounded.

$a_{n+1} - a_n = (n+1)/(n+2) - n/(n+1) = [(n+1)^2 - n(n+2)] / [(n+1)(n+2)] =$ **$1 / [(n+1)(n+2)] > 0$**, so it is increasing.

Since $n < n + 1$, $a_n < 1$, and $a_n \geq 1/2 > 0$. So it is bounded: $1/2 \leq a_n < 1$.

1.3 The limit of a sequence

THE IDEA IN PLAIN ENGLISH

If the terms of a sequence **settle down and get closer and closer to one fixed number L**, we say the sequence **converges to L** and write **lim a_n = L**. If there is no such number, the sequence **diverges**.

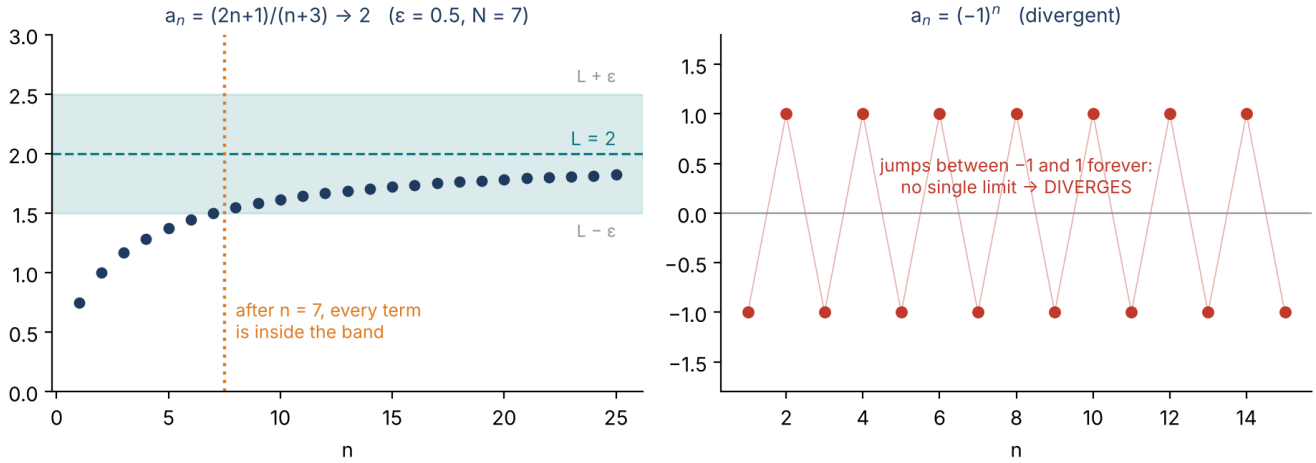
There are two kinds of divergence: **going off to ±∞** (like $n^2 \to \infty$) and **refusing to settle** (like $(-1)^n$, which keeps bouncing between −1 and 1).

The formal (ε–N) definition

lim a_n = L means for every ε > 0 there is an N such that $|a_n - L| < \varepsilon$ for all n > N

THE IDEA IN PLAIN ENGLISH

Read it as a game. You pick any tiny tolerance ε (how close you demand). I must be able to find a cut-off N so that **after that point every single term** is within ε of L. If I can always win, the limit is L. In the picture below, ε = 0.5 gives a band from 1.5 to 2.5 around L = 2, and after n = 7 every term stays inside the band.



Left: a convergent sequence trapped in the ε-band. Right: a divergent sequence that oscillates.

WORKED EXAMPLE

Prove from the definition that (2n+1)/(n+3) → 2.

Work backwards: $|a_n - 2| = |(2n+1 - 2n - 6)/(n+3)| =$ **5/(n+3)**. We want this < ε. Since 5/(n+3) < 5/n, it is enough that 5/n < ε, i.e. n > 5/ε.

Proof: Given ε > 0, choose N = 5/ε. Then for n > N, $|a_n - 2| < 5/n < \varepsilon$. Hence $a_n \to 2$. ∎

JOT THIS DOWN

Sequence = ordered list $a_1, a_2, \ldots$; **general term** a_n. Explicit formula vs recursive rule.

Increasing: $a_{n+1} \geq a_n$. **Decreasing:** $a_{n+1} \leq a_n$. **Bounded:** $|a_n| \leq K$.

Convergent: terms approach one number L. Else **divergent** (to ±∞, or oscillating).

ε–N: $\forall \varepsilon > 0\ \exists N: n > N \Rightarrow |a_n - L| < \varepsilon$. Method: simplify $|a_n - L|$, then solve for n.

1.4 Finding limits: the toolkit

Limit laws

If $a_n \to A$ and $b_n \to B$, then: $\mathbf{(a_n \pm b_n) \to A \pm B}$; $\mathbf{c \cdot a_n \to cA}$; $\mathbf{a_n b_n \to AB}$; $\mathbf{a_n / b_n \to A/B}$ (if $B \neq 0$); $\mathbf{a_n^p \to A^p}$. Also, if f is continuous at A then $\mathbf{f(a_n) \to f(A)}$.

Standard limits you must know

Sequence	Limit	Condition
c (a constant)	c	—
$1/n^p$	0	$p > 0$
r^n	0 if $\lvert r\rvert < 1$; 1 if $r = 1$; ∞ if $r > 1$; diverges if $r \leq -1$	geometric sequence
$n^{1/n}$ and $c^{1/n}$	1	$c > 0$
$(1 + x/n)^n$	e^x	any real x ($x = 1$ gives e)
$x^n/n!$	0	any real x
$(\ln n)/n^p$	0	$p > 0$

THE IDEA IN PLAIN ENGLISH

Growth-rate ladder (slowest to fastest): $\mathbf{\ln n \ll n^p \ll a^n\ (a > 1) \ll n! \ll n^n}$. In a fraction, the term higher up the ladder wins. So $n^2/2^n \to 0$, $2^n/n! \to 0$, $n!/n^n \to 0$, but $n!/10^n \to \infty$.

Worked limits

WORKED EXAMPLE

(a) Divide by the highest power. $(3n^2 + 2n)/(n^2 - 1) = (3 + 2/n)/(1 - 1/n^2) \to \mathbf{3}$.

(b) Rationalise a difference of roots. $\sqrt{(n^2+n)} - n = n/(\sqrt{(n^2+n)} + n) = 1/(\sqrt{(1+1/n)} + 1) \to \mathbf{1/2}$.

(c) Substitution. $n \sin(1/n)$: let $x = 1/n \to 0$, giving $\sin x / x \to \mathbf{1}$.

(d) L'Hôpital (replace n by x first): $(\ln n)/n \to (1/x)/1 \to \mathbf{0}$.

(e) Special limit. $(1 + 2/n)^n \to \mathbf{e^2}$.

(f) Squeeze (sandwich). $-1/n \leq (\sin n)/n \leq 1/n$ and both sides $\to 0$, so $(\sin n)/n \to \mathbf{0}$.

(g) Squeeze on a product. $n!/n^n = (1/n)(2/n)(3/n)\ldots(n/n) \leq 1/n$, so $0 < n!/n^n \leq 1/n \to \mathbf{0}$.

(h) Roots of sums. $3 \leq (2^n + 3^n)^{1/n} \leq (2 \cdot 3^n)^{1/n} = 3 \cdot 2^{1/n} \to 3$. By squeeze, the limit is **3**.

THE IDEA IN PLAIN ENGLISH

Squeeze (Sandwich) Theorem: if $b_n \leq a_n \leq c_n$ and $b_n \to L$ and $c_n \to L$, then $a_n \to L$. Use it when the middle sequence is awkward but can be trapped between two easy ones.

Absolute-value shortcut: if $|a_n| \to 0$ then $a_n \to 0$. That is how $(-1)^n/n \to 0$.

1.5 Monotone convergence and recursive sequences

THE IDEA IN PLAIN ENGLISH

Monotone Convergence Theorem: every bounded monotonic sequence converges. Picture a ball that only rolls one way (monotonic) but is walled in (bounded). It has to stop somewhere. Increasing and bounded above, or decreasing and bounded below, is enough.

Other facts: a limit, if it exists, is **unique**. Every convergent sequence is **bounded** (but bounded does not mean convergent: $(-1)^n$). If a sequence converges, **every subsequence** converges to the same limit, so if two subsequences have different limits, the sequence **diverges** (this is the quick proof for $(-1)^n$ and for $(-1)^n \cdot n/(n+1)$).

WORKED EXAMPLE

Recursive sequence. $a_1 = \sqrt{2}$, $a_{n+1} = \sqrt{(2 + a_n)}$. Show it converges and find the limit.

Step 1: bounded above by 2 (induction). $a_1 = \sqrt{2} < 2$. If $a_n < 2$, then $a_{n+1} = \sqrt{(2 + a_n)} < \sqrt{4} = 2$.

Step 2: increasing. $a_{n+1}^2 - a_n^2 = 2 + a_n - a_n^2 = (2 - a_n)(1 + a_n) > 0$ because $a_n < 2$. So $a_{n+1} > a_n$.

Step 3: conclude. Increasing and bounded above ⇒ converges to some L.

Step 4: find L. Let $n \to \infty$ on both sides of the rule: $L = \sqrt{(2 + L)} \Rightarrow L^2 - L - 2 = 0 \Rightarrow (L - 2)(L + 1) = 0$. Terms are positive so $L \neq -1$. **L = 2.**

WATCH OUT: EXAM TRAP

Never find L from "L = f(L)" before proving the sequence converges. The trick assumes the limit exists. Prove it first (monotone + bounded), then solve for L, then reject any root that does not fit (negative, or outside the bounds).

JOT THIS DOWN

Limit laws: limits of sums, products, quotients (denominator limit ≠ 0) are sums, products, quotients of limits.

Standard: $1/n^p \to 0$; $r^n \to 0$ if $|r| < 1$; $n^{1/n} \to 1$; $(1 + x/n)^n \to e^x$; $x^n/n! \to 0$.

Growth: $\ln n \ll n^p \ll a^n \ll n! \ll n^n$.

Tools: divide by highest power; rationalise roots; L'Hôpital (use x for n); squeeze; $|a_n| \to 0 \Rightarrow a_n \to 0$.

Monotone Convergence: bounded + monotone ⇒ convergent. Recursive: prove convergence, then solve L = f(L).

Divergence by subsequences: two subsequences with different limits ⇒ diverges.

PART 2 | Progressions and Finite Sums

2.1 Arithmetic progression (AP)

THE IDEA IN PLAIN ENGLISH

An **AP** is a sequence where you **add the same number every time**. That number is the **common difference d**. Examples: 3, 7, 11, 15, … (d = 4) and 20, 17, 14, … (d = −3). Climbing a staircase with equal steps is an AP.

$$T_n = a + (n-1)d \quad S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$$

- **a** = first term, **d** = common difference = $T_{n+1} - T_n$, **l** = last term, **n** = number of terms.
- **Why $S_n = \frac{n}{2}(a + l)$:** write the sum forwards and backwards and add: each of the n vertical pairs adds to (a + l), giving $2S_n = n(a + l)$.
- **Useful facts:** $T_n = S_n - S_{n-1}$. Three numbers in AP: **a − d, a, a + d**. Three consecutive terms: the middle one is the average of its neighbours.

WORKED EXAMPLE

Example 1. Sum of the first 20 terms of 3, 7, 11, …: a = 3, d = 4. $S_{20} = (20/2)[6 + 19(4)] = 10 \times 82 =$ **820**.

Example 2. How many terms of 2, 5, 8, … give a sum of 345? $(n/2)[4 + 3(n-1)] = 345 \Rightarrow n(3n + 1) = 690 \Rightarrow 3n^2 + n - 690 = 0 \Rightarrow n = (-1 \pm 91)/6$. Reject the negative root: **n = 15**.

Example 3. $T_3 = 7$ and $T_8 = 22$. Subtract: 5d = 15 so d = 3, then a = 7 − 6 = 1. $T_n = 3n - 2$ and $S_{10} = 5(2 + 27) =$ **145**.

Example 4. Sum of all multiples of 7 between 1 and 100: 7, 14, …, 98. Number of terms 98 = 7n so n = 14. $S = (14/2)(7 + 98) =$ **735**.

2.2 Geometric progression (GP)

THE IDEA IN PLAIN ENGLISH

A **GP** is a sequence where you **multiply by the same number every time**, the **common ratio r**. Examples: 2, 6, 18, 54, … (r = 3) and 8, 4, 2, 1, … (r = 1/2). Money growing with compound interest is a GP.

$$T_n = a\,r^{n-1} \quad S_n = \frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r} \; (r \neq 1)$$

$$\text{Sum to infinity: } S_\infty = \frac{a}{1 - r} \text{ provided } |r| < 1$$

- **$r = T_{n+1}/T_n$.** Use the form with the positive denominator: $(r^n - 1)/(r - 1)$ when r > 1; $(1 - r^n)/(1 - r)$ when r < 1.
- **Why S_∞ exists only when $|r| < 1$:** then $r^n \to 0$, so $S_n \to a/(1 - r)$. If $|r| \geq 1$ the terms do not shrink and the sum does not settle.
- **Three numbers in GP:** a/r, a, ar.

WORKED EXAMPLE

Example 1. S_8 of 2, 6, 18, …: a = 2, r = 3. $S_8 = 2(3^8 - 1)/2 = 6561 - 1 =$ **6560**.

Example 2. $T_2 = 6$, $T_5 = 48$. $T_5/T_2 = r^3 = 8 \Rightarrow r = 2$, $a = 3$. $S_6 = 3(2^6 - 1) = \mathbf{189}$.

Example 3. $6 + 2 + 2/3 + \ldots$: $a = 6$, $r = 1/3$. $S_\infty = 6/(1 - 1/3) = \mathbf{9}$.

Recurring decimals as fractions. $0.272727\ldots = 0.27 + 0.0027 + \ldots = a/(1 - r)$ with $a = 0.27$, $r = 0.01$: $0.27/0.99 = \mathbf{3/11}$. And $0.1666\ldots = 1/10 + (0.06 + 0.006 + \ldots) = 1/10 + 0.06/0.9 = 1/10 + 1/15 = \mathbf{1/6}$.

2.3 Harmonic progression (HP) and the three means

THE IDEA IN PLAIN ENGLISH

A **harmonic progression** is a sequence whose **reciprocals form an AP**. So 1/2, 1/5, 1/8, … is an HP because 2, 5, 8, … is an AP. There is no direct sum formula: **flip it into an AP, work there, flip back**. Example: the 10th term of 1/2, 1/5, 1/8, … = $1/(2 + 9\times3)$ = **1/29**.

Mean of a and b	Formula	Idea
Arithmetic (AM)	$(a + b)/2$	The middle term of an AP
Geometric (GM)	$\sqrt{(ab)}$	The middle term of a GP
Harmonic (HM)	$2ab/(a + b)$	The middle term of an HP

- **AM ≥ GM ≥ HM** for positive numbers, with equality only when $a = b$. Also **$GM^2 = AM \times HM$**.
- Example: $a = 4$, $b = 9$: AM = 6.5, GM = 6, HM = 72/13 ≈ 5.54. Check $GM^2 = 36 = 6.5 \times 72/13$ ✓.
- **Inserting k arithmetic means** between a and b makes an AP with $k + 2$ terms, so $d = (b - a)/(k + 1)$. Insert 4 AMs between 3 and 23: $d = 20/5 = 4 \rightarrow$ **7, 11, 15, 19**.

2.4 Sigma notation and standard sums

THE IDEA IN PLAIN ENGLISH

Σ (sigma) is a shorthand for "add up". $\Sigma_{k=1}^{n} a_k$ means $a_1 + a_2 + \ldots + a_n$: *k* is the counter, it starts at the bottom value and runs to the top value.

- **Linearity:** $\Sigma(c\cdot a_k) = c\cdot\Sigma a_k$; $\Sigma(a_k \pm b_k) = \Sigma a_k \pm \Sigma b_k$; $\mathbf{\Sigma_{k=1}^{n} c = nc}$ (adding a constant n times).
- **Shifting the index:** $\Sigma_{k=1}^{n} a_k = \Sigma_{j=0}^{n-1} a_{j+1}$. Count of terms from m to n = $n - m + 1$.
- **Not allowed:** $\Sigma(a_k b_k) \neq (\Sigma a_k)(\Sigma b_k)$.

$$\mathbf{\Sigma k = n(n+1)/2 \quad \Sigma k^2 = n(n+1)(2n+1)/6 \quad \Sigma k^3 = [n(n+1)/2]^2 \ (k = 1 \text{ to } n)}$$

WORKED EXAMPLE

Example. $\Sigma_{k=1}^{10} (2k^2 - 3k + 1) = 2(385) - 3(55) + 10 = 770 - 165 + 10 = \mathbf{615}$ (using $\Sigma k^2 = 385$, $\Sigma k = 55$, and $\Sigma 1 = 10$).

Sum of the first n odd numbers. $\Sigma(2k - 1) = 2\cdot n(n+1)/2 - n = n^2 + n - n = \mathbf{n^2}$.

Telescoping sums

THE IDEA IN PLAIN ENGLISH

Sometimes you can write each term as a **difference** so that the middle terms cancel like the sections of a folding telescope, leaving only the first and last pieces. The usual tool is **partial fractions**.

WORKED EXAMPLE

$\Sigma_{k=1}^{n}$ 1/[k(k+1)]. Since 1/[k(k+1)] = 1/k − 1/(k+1):

(1 − 1/2) + (1/2 − 1/3) + … + (1/n − 1/(n+1)) = **1 − 1/(n+1) = n/(n+1)**.

$\Sigma_{k=1}^{n}$ (√(k+1) − √k) = √(n+1) − 1 (everything cancels except the ends).

Proof by induction (a standard exam question)

WORKED EXAMPLE

Prove 1 + 2 + … + n = n(n+1)/2 for all n ≥ 1.

Base case: n = 1: LHS = 1, RHS = 1·2/2 = 1 ✓.

Inductive step: assume true for n = k: 1 + … + k = k(k+1)/2. Add the next term (k+1): k(k+1)/2 + (k+1) = (k+1)(k/2 + 1) = (k+1)(k+2)/2, which is the formula for n = k + 1 ✓.

By induction the formula holds for all n ≥ 1. ∎

JOT THIS DOWN

AP: $T_n = a + (n-1)d$; $S_n = (n/2)[2a + (n-1)d] = (n/2)(a + l)$.

GP: $T_n = ar^{n-1}$; $S_n = a(r^n-1)/(r-1)$; **S_∞ = a/(1−r), |r| < 1.**

HP: reciprocals form an AP. **AM = (a+b)/2, GM = √(ab), HM = 2ab/(a+b); AM ≥ GM ≥ HM; GM^2 = AM·HM.**

Σk = n(n+1)/2; Σk^2 = n(n+1)(2n+1)/6; Σk^3 = [n(n+1)/2]²; Σc = nc.

Telescoping: split into a difference (partial fractions), cancel the middle. 1/[k(k+1)] = 1/k − 1/(k+1).

Induction: base case, assume for k, prove for k + 1.

PART 3 | Infinite Series: the Basics

3.1 From a sequence to a series

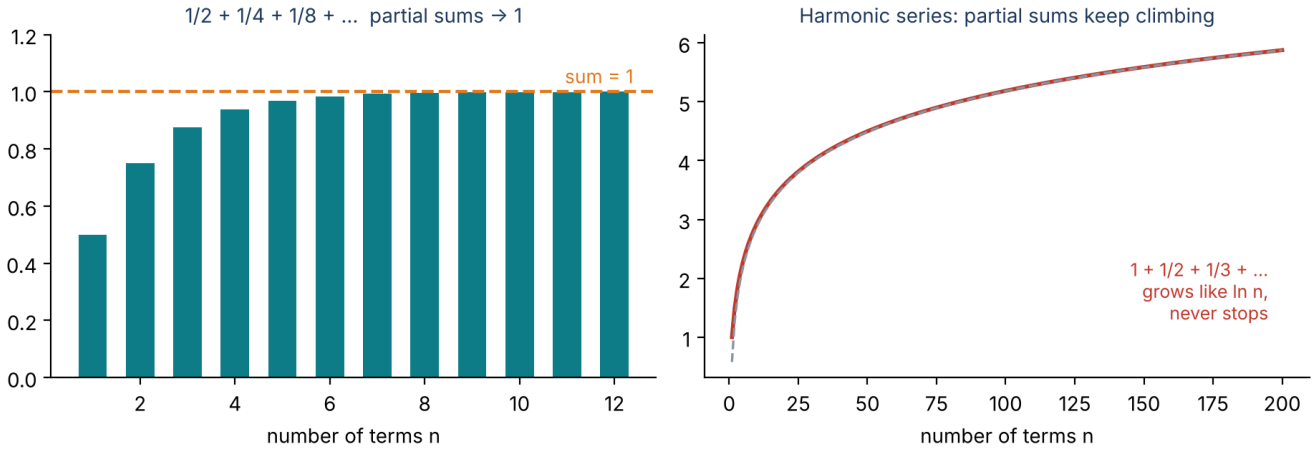
THE IDEA IN PLAIN ENGLISH

A **series** is what you get when you **add up the terms of a sequence**. A sequence is a list; a series is the total of that list.

Adding infinitely many numbers sounds impossible, so we do it **step by step**. Build the **partial sums**: $S_1 = a_1$, $S_2 = a_1 + a_2$, $S_3 = a_1 + a_2 + a_3$, … These partial sums form a new sequence $\{S_n\}$. **If the partial sums settle down to a number S, we say the series converges to S.** If they do not, it diverges.

$$\Sigma_{n=1}^{\infty} a_n = S \text{ means } S_n = a_1 + \ldots + a_n \to S$$

- **Link between terms and partial sums:** $a_n = S_n - S_{n-1}$.
- **Example:** if $S_n = n/(n+1)$, then $S_n \to 1$, so the series sums to 1, and $a_n = n/(n+1) - (n-1)/n =$ **1/[n(n+1)]**.
- A series that does not converge is **divergent**. A divergent series has no sum (it may go to ∞, −∞ or oscillate).



Left: partial sums of 1/2 + 1/4 + 1/8 + ... creep up to 1 (converges). Right: partial sums of the harmonic series never stop growing (diverges).

3.2 The geometric series

$$\sum_{n=0}^{\infty} a\,r^n = a/(1-r) \text{ if } |r|<1; \text{ diverges if } |r|\geq 1$$

THE IDEA IN PLAIN ENGLISH

General rule that survives any starting index: sum = **(first term) / (1 − r)**. Always identify the first term and the ratio r = (next term)/(this term).

WORKED EXAMPLE

(a) $\sum_{n=1}^{\infty} 3(1/4)^n$: first term 3/4, r = 1/4. Sum = (3/4)/(3/4) = **1**.

(b) $\sum_{n=0}^{\infty} (-2/3)^n$: first term 1, r = −2/3. Sum = 1/(1 + 2/3) = **3/5**.

(c) $\sum_{n=2}^{\infty} (1/3)^n$: first term 1/9, r = 1/3. Sum = (1/9)/(2/3) = **1/6**.

(d) $\sum_{n=1}^{\infty} 2^{n+1}/3^n$: each term equals $2\cdot(2/3)^n$, so the first term is 4/3 and r = 2/3. Sum = (4/3)/(1/3) = **4**.

WATCH OUT: EXAM TRAP

Check the starting index. $\sum_{n=1}^{\infty} r^n = r/(1-r)$, **not** 1/(1 − r) (that is for n = 0). Writing "first term over (1 − r)" avoids this slip.

3.3 Telescoping series

WORKED EXAMPLE

$\sum_{n=1}^{\infty}$ 1/[n(n+2)]. Partial fractions: 1/[n(n+2)] = (1/2)(1/n − 1/(n+2)).

S_n = (1/2)[(1 − 1/3) + (1/2 − 1/4) + (1/3 − 1/5) + ... + (1/n − 1/(n+2))] = (1/2)[1 + 1/2 − 1/(n+1) − 1/(n+2)].

As n → ∞ the last two fractions vanish: sum = (1/2)(3/2) = **3/4**.

Note that here two terms survive at each end, because the gap is 2.

3.4 The nth-term (divergence) test

THE IDEA IN PLAIN ENGLISH

If a series converges, its terms must shrink to 0. (If you keep adding chunks that do not get small, the total cannot settle.) So:

If lim $a_n \neq 0$ (or does not exist), the series DIVERGES.

WATCH OUT: EXAM TRAP

The test only works one way. If $a_n \to 0$ the test says **nothing**: the series could converge or diverge. The classic example is the **harmonic series Σ 1/n**: its terms go to 0, yet it diverges. "Terms go to 0" is **necessary but not sufficient** for convergence.

WORKED EXAMPLE

Σ n/(2n+1): terms → 1/2 ≠ 0, so it **diverges**.

Σ $(1 + 1/n)^n$: terms → e ≠ 0, so it **diverges**.

Σ $(-1)^n$: terms do not → 0, so it **diverges** (partial sums bounce between −1 and 0).

3.5 The harmonic series and p-series

THE IDEA IN PLAIN ENGLISH

Why 1 + 1/2 + 1/3 + 1/4 + … diverges (the grouping argument): 1 + 1/2 + (1/3 + 1/4) + (1/5 + … + 1/8) + (1/9 + … + 1/16) + …

Each bracket is at least 1/2 (e.g. 1/3 + 1/4 > 1/4 + 1/4 = 1/2; four terms 1/5…1/8 each ≥ 1/8, total ≥ 1/2). Infinitely many brackets each worth at least 1/2 add up to ∞. It grows very slowly, like ln n, but it never stops.

p-series: $\Sigma_{n=1}^{\infty} 1/n^p$ converges if p > 1, diverges if p ≤ 1

- Examples: $\Sigma 1/n^2$ (p = 2) converges (its sum is $\pi^2/6$, a famous result of Euler); $\Sigma 1/n^{3/2}$ converges; $\Sigma 1/\sqrt{n}$ (p = 1/2) diverges; $\Sigma 1/n$ (p = 1) diverges.
- The p-series is the yardstick for the comparison tests in Part 4.

3.6 Basic properties

- **Constant multiples:** $\Sigma c \cdot a_n$ converges iff Σa_n converges (c ≠ 0), and the sum is c times as big.
- **Sum of two convergent series** converges: $\Sigma(a_n + b_n) = \Sigma a_n + \Sigma b_n$.
- **Convergent + divergent = divergent.** (But divergent + divergent can converge, e.g. Σ1/n + Σ(−1/n).)
- **Changing finitely many terms** (or deleting the first few) does **not** change whether the series converges; it only changes the sum.

JOT THIS DOWN

Series = sum of a sequence. **Partial sum** $S_n = a_1 + \ldots + a_n$. Series converges to S iff $S_n \to S$. $a_n = S_n - S_{n-1}$.

Geometric: Σ ar^n = (first term)/(1 − r) if |r| < 1; diverges if |r| ≥ 1.

nth-term test: $a_n \not\to 0 \Rightarrow$ diverges. $a_n \to 0 \Rightarrow$ inconclusive (harmonic series!).

Harmonic Σ1/n diverges. p-series $\Sigma 1/n^p$: converges iff p > 1.

Telescoping: partial fractions → cancel → take limit of S_n. $\Sigma 1/[n(n+2)] = 3/4$.

Convergent + divergent ⇒ divergent. Changing finitely many terms doesn't affect convergence.

PART 4 | Convergence Tests for Series

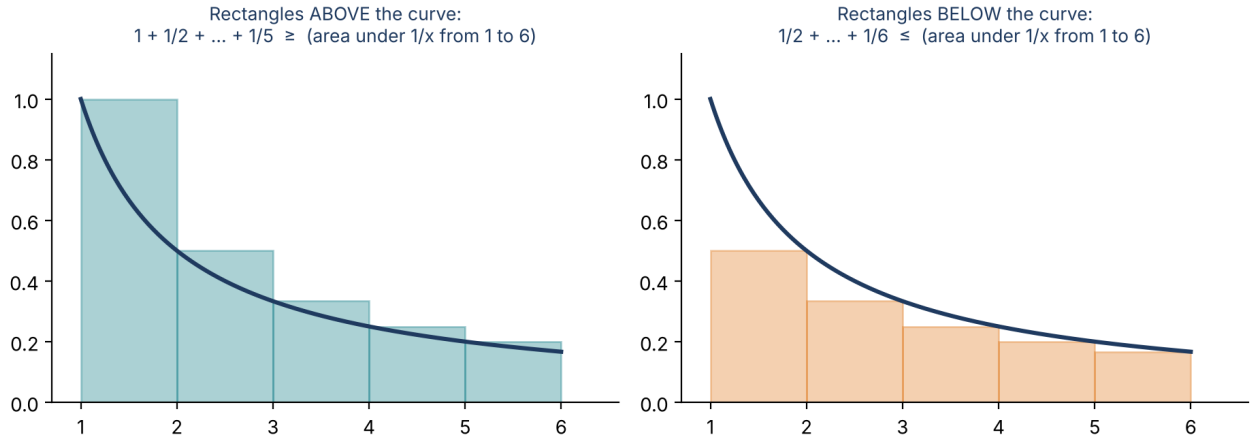
Most series cannot be summed exactly, so we ask the cheaper question: **does it converge at all?** Each test below is a tool for a certain shape of series. Learning *which tool for which shape* is the skill examiners test.

4.1 Integral test

THE IDEA IN PLAIN ENGLISH

Compare the sum with an **area**. If $a_n = f(n)$ where f is **positive, continuous and decreasing** for $x \geq 1$, then **Σa_n and $\int_1^\infty f(x)dx$ converge together or diverge together.**

Why: rectangles of width 1 and height a_n sit just above (or below) the curve, so the sum and the area can only differ by a bounded amount.



The rectangles above the curve show Σ ≥ area (so if the area is infinite the sum diverges); those below show Σ ≤ area + 1st term (so if the area is finite the sum converges).

WORKED EXAMPLE

Σ 1/(n ln n), n ≥ 2. $\int dx/(x \ln x) = \ln(\ln x) \to \infty$. **Diverges.**

Σ n e^{-n^2}. $f(x) = x\,e^{-x^2}$ is decreasing for $x \geq 1$. $\int_1^\infty x\,e^{-x^2}\,dx = [-\tfrac{1}{2}e^{-x^2}]_1^\infty = \tfrac{1}{2}e^{-1}$ (finite). **Converges.**

Proof of the p-series rule. $\int_1^\infty x^{-p}\,dx = 1/(p-1)$ when $p > 1$ (finite), and infinite when $p \leq 1$.

WATCH OUT: EXAM TRAP

Check all three conditions (positive, continuous, decreasing) before using the test. The integral's **value is not the sum**; it only tells you convergence. For a remainder estimate: $\int_{n+1}^\infty f \leq R_n \leq \int_n^\infty f$.

4.2 Comparison test (for positive terms)

THE IDEA IN PLAIN ENGLISH

Squeeze your series between a known one. If $\mathbf{0 \le a_n \le b_n}$:

• $\mathbf{\Sigma b_n}$ **converges** ⇒ $\mathbf{\Sigma a_n}$ **converges** (smaller than a convergent series). • $\mathbf{\Sigma a_n}$ **diverges** ⇒ $\mathbf{\Sigma b_n}$ **diverges** (bigger than a divergent series).

Memory: **“Smaller than a finite thing is finite. Bigger than an infinite thing is infinite.”** Good yardsticks: geometric series and p-series.

WORKED EXAMPLE

Σ 1/(2^n + 1): $1/(2^n + 1) < 1/2^n$ and $\Sigma(1/2)^n$ converges (geometric). **Converges.**

Σ (ln n)/n: for $n \ge 3$, $\ln n > 1$, so $(\ln n)/n > 1/n$, and $\Sigma 1/n$ diverges. **Diverges.**

Σ 1/(n² + n + 1): $< 1/n^2$ and $\Sigma 1/n^2$ converges. **Converges.**

WATCH OUT: EXAM TRAP

The inequality must point the right way. $\Sigma\ 1/(n^2 - 1)$ is *bigger* than $\Sigma\ 1/n^2$, and “bigger than a convergent series” proves nothing. Use the limit comparison test instead.

4.3 Limit comparison test

THE IDEA IN PLAIN ENGLISH

For **positive** a_n, b_n, if $\mathbf{\lim a_n/b_n = c}$ **with** $\mathbf{0 < c < \infty}$, then Σa_n and Σb_n **do the same thing** (both converge or both diverge). It says: for large n the two series are basically proportional.

How to choose $\mathbf{b_n}$: keep only the **highest-power term** on top and bottom of a_n and simplify. That gives a p-series or geometric series.

WORKED EXAMPLE

Σ (3n + 1)/(n³ + 2n): behaves like $3n/n^3 = 3/n^2$. Take $b_n = 1/n^2$: $a_n/b_n \to 3$. $\Sigma 1/n^2$ converges ⇒ **converges.**

Σ sin(1/n): positive for $n \ge 1$; take $b_n = 1/n$: $\sin(1/n)/(1/n) \to 1$. $\Sigma 1/n$ diverges ⇒ **diverges.**

Σ 1/√(n² + 1): like 1/n. Ratio → 1 ⇒ **diverges.**

Σ 1/(n² − 1), n ≥ 2: $b_n = 1/n^2$, ratio → 1 ⇒ **converges.**

4.4 Ratio test

$L = \lim |a_{n+1} / a_n|$ L < 1: converges absolutely L > 1 (or ∞): diverges L = 1: no information

THE IDEA IN PLAIN ENGLISH

If successive terms shrink by a factor $L < 1$ each time, the series behaves like a geometric series with ratio L, so it converges. If the factor is above 1, the terms grow and the series diverges. **Best for factorials, exponentials (a^n), and products.**

WORKED EXAMPLE

$\Sigma\ 2^n/n!$: ratio $= 2/(n+1) \to 0 < 1$. **Converges** (its sum is $e^2 - 1$ when started at n = 1).

Σ n!/n^n: ratio = $(n+1)!n^n/[(n+1)^{n+1}n!] = (n/(n+1))^n = 1/(1 + 1/n)^n \to 1/e < 1$. **Converges.**

Σ $n^2/2^n$: ratio = $(n+1)^2/(2n^2) \to 1/2$. **Converges.**

Σ $3^n/n^2$: ratio = $3n^2/(n+1)^2 \to 3 > 1$. **Diverges.**

Σ $n!/3^n$: ratio = $(n+1)/3 \to \infty$. **Diverges.**

WATCH OUT: EXAM TRAP

L = 1 means the test failed, not that the series diverges. For every p-series the ratio → 1, so the ratio test is useless on $\Sigma 1/n^p$ and on any series that looks like a rational function of n. Switch to limit comparison.

4.5 Root test

$L = \lim |a_n|^{1/n}$ L < 1: converges absolutely L > 1 (or ∞): diverges L = 1: no information

THE IDEA IN PLAIN ENGLISH

Same decision rule as the ratio test, but using the nth root. **Best when the whole term is raised to the power n** (so the root removes the power).

WORKED EXAMPLE

Σ $((2n+1)/(3n+2))^n$: nth root = $(2n+1)/(3n+2) \to 2/3 < 1$. **Converges.**

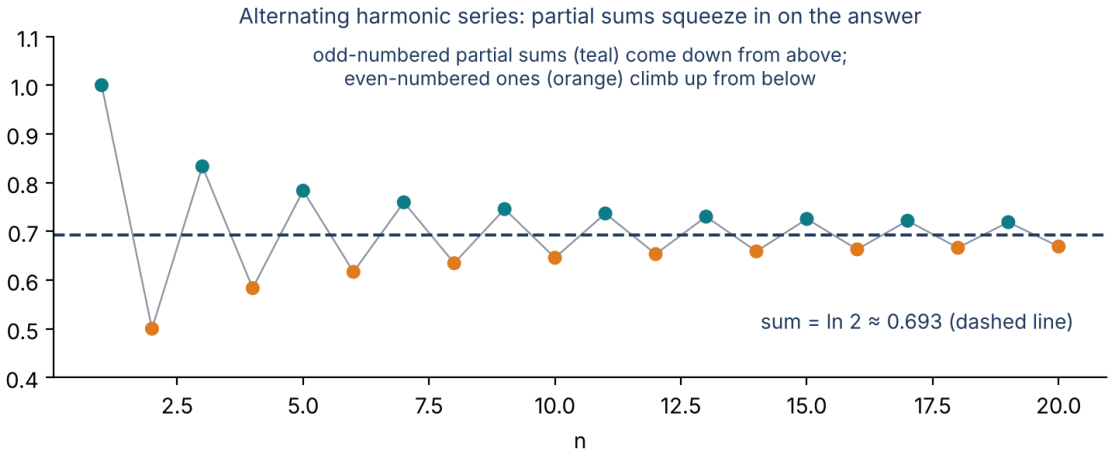
Σ $(n/(n+1))^{n^2}$: nth root = $(n/(n+1))^n \to 1/e < 1$. **Converges.**

4.6 Alternating series test (Leibniz)

THE IDEA IN PLAIN ENGLISH

An **alternating series** has terms that switch sign: $\Sigma(-1)^{n-1}b_n$ with $b_n > 0$. **If (i) b_n is decreasing (at least eventually) and (ii) $b_n \to 0$, then the series converges.**

Picture walking along a line: step right a big distance, left a slightly smaller one, right smaller again... each step is shorter, so you spiral in on a final position.



Partial sums of 1 − 1/2 + 1/3 − … swing above and below ln 2 and get closer each time.

THE IDEA IN PLAIN ENGLISH

Error bound: if you stop after n terms, the error is **no bigger than the first omitted term**: $|S - S_n| \le b_{n+1}$, and the true sum lies between S_n and S_{n+1}.

WORKED EXAMPLE

$\Sigma(-1)^{n+1}/n$ (alternating harmonic): 1/n decreasing and → 0 ⇒ **converges** (to ln 2).

$\Sigma(-1)^n\, n/(n+1)$: terms → ±1, not 0, so the nth-term test says **diverges** (condition (ii) fails).

Estimate $\Sigma(-1)^{n+1}/n!$ within 0.01. Need $b_{n+1} = 1/(n+1)! < 0.01$: $1/5! = 0.0083 < 0.01$, so use n = 4 terms: $1 - 1/2 + 1/6 - 1/24 =$ **0.625** (true sum $1 - 1/e \approx 0.632$, error 0.007 ✓).

How many terms of $\Sigma(-1)^{n+1}/n$ give error < 0.01? Need $1/(n+1) < 0.01 \Rightarrow n + 1 > 100 \Rightarrow$ **n = 100 terms**.

4.7 Absolute and conditional convergence

THE IDEA IN PLAIN ENGLISH

For a series with mixed signs, ask two questions.

Q1: does $\Sigma|a_n|$ converge? If yes the series is **absolutely convergent**, and that **guarantees** Σa_n converges too. Absolute convergence is the strongest, safest kind.

Q2: if $\Sigma|a_n|$ diverges but Σa_n still converges (thanks to the cancelling signs), it is **conditionally convergent**.

Series	$\Sigma\|a_n\|$	Verdict
$\Sigma(-1)^n/n^2$	$\Sigma 1/n^2$ converges	**Absolutely** convergent
$\Sigma(-1)^{n+1}/n$	$\Sigma 1/n$ diverges, but AST says converges	**Conditionally** convergent
$\Sigma(-1)^n/\sqrt{n}$	$\Sigma 1/\sqrt{n}$ diverges, AST converges	**Conditionally** convergent
$\Sigma \sin n / n^2$	$\lvert\sin n/n^2\rvert \le 1/n^2$, converges	**Absolutely** convergent
$\Sigma(-1)^n \ln n / n$	$\Sigma \ln n/n$ diverges; AST works ($\ln n/n$ decreasing for $n \ge 3$, → 0)	**Conditionally** convergent

BEYOND THE SLIDE (still examinable)

Rearranging terms: an absolutely convergent series can be reordered without changing its sum. A conditionally convergent series cannot: by rearranging its terms you can make it add up to *any* number you like (Riemann's rearrangement theorem).

4.8 Which test do I use? (decision guide)

Look at the series. Does it…	Use	Watch out
Have terms not tending to 0?	**nth-term test** → diverges	Does not prove convergence
Look like ar^n? Or $\Sigma 1/n^p$?	**Geometric / p-series** rule	Check start index; p > 1
Split into a difference (partial fractions)?	**Telescoping**: find S_n	Count surviving terms
Contain **n!**, a^n, or products of terms?	**Ratio test**	L = 1 is inconclusive
Have the whole term to the power **n**?	**Root test**	L = 1 is inconclusive
Look like a rational / algebraic function of n?	**Limit comparison** with p-series (highest powers)	Terms must be positive
Compare easily with a known inequality?	**Direct comparison**	Inequality direction!
Have f(x) easy to integrate (e.g. ln, e^{-x})?	**Integral test**	Positive, continuous, decreasing
Alternate in sign?	**Alternating series test**, then check absolute convergence	Needs b_n decreasing and → 0

JOT THIS DOWN

Integral: f positive, continuous, decreasing: $\Sigma f(n)$ and $\int f$ converge together.

Comparison: $0 \le a_n \le b_n$; Σb converges ⇒ Σa converges; Σa diverges ⇒ Σb diverges.

Limit comparison: $\lim a_n/b_n = c$, $0 < c < \infty$ ⇒ same behaviour. Choose b_n from highest powers.

Ratio: $L = \lim|a_{n+1}/a_n|$; <1 converges, >1 diverges, =1 inconclusive (factorials, a^n).

Root: $L = \lim|a_n|^{1/n}$; same rule (term raised to n).

Alternating: b_n decreasing and → 0 ⇒ converges; error ≤ first omitted term.

Absolute: $\Sigma|a_n|$ converges ⇒ Σa_n converges. **Conditional**: Σa_n converges but $\Sigma|a_n|$ diverges.

PART 5 | Power Series

5.1 What is a power series?

THE IDEA IN PLAIN ENGLISH

A **power series** is a series that contains a variable x. It is like a polynomial of **infinite degree**:

$\mathbf{\Sigma_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots}$ (a = the **centre**, c_n = the **coefficients**)

For each value of x you plug in, you get an ordinary series of numbers that either converges or diverges. So the real question is: **for which x does it converge?** The answer is always an interval centred at a.

THE IDEA IN PLAIN ENGLISH

Exactly one of three things happens:

(1) It converges **only at x = a** (radius of convergence **R = 0**).

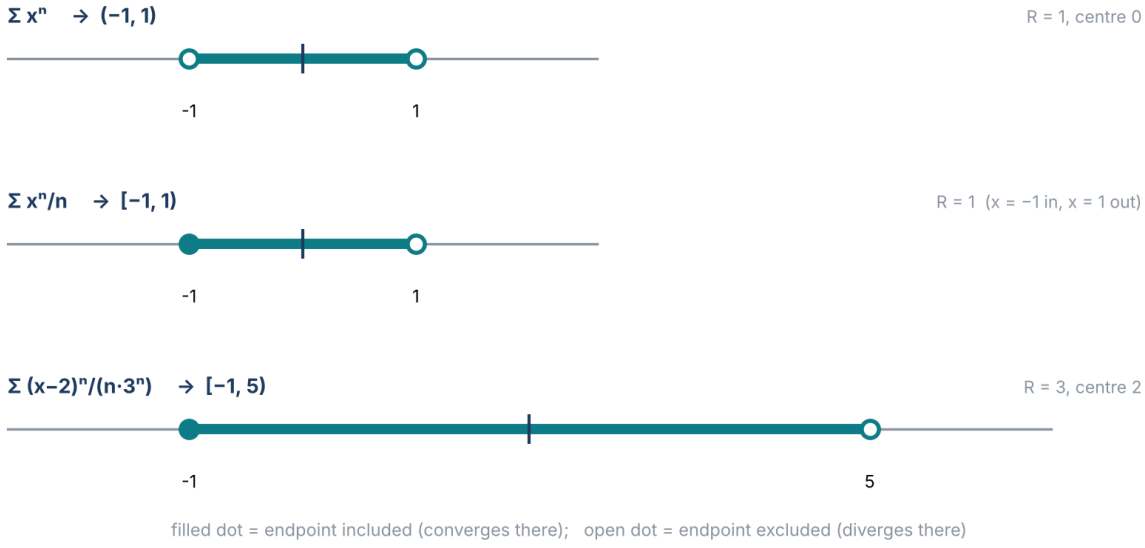
(2) It converges **for every x** (**R = ∞**).

(3) There is a number **R > 0**: it converges (absolutely) when **|x − a| < R** and diverges when |x − a| > R. **At the two endpoints x = a ± R anything can happen, so test them separately.**

The set of x where it converges is the **interval of convergence**: R is the **radius**, (a − R, a + R) the open interval, then add whichever endpoints converge.

The standard procedure

- **Step 1.** Apply the **ratio test** to the whole term: $L = \lim |c_{n+1}(x-a)^{n+1} / c_n(x-a)^n| = |x-a| \cdot K$.
- **Step 2.** Solve $|x-a| \cdot K < 1 \Rightarrow |x-a| < 1/K$. So **R = 1/K** ($R = \infty$ if $K = 0$; $R = 0$ if $K = \infty$). Equivalent: $R = \lim |c_n/c_{n+1}|$.
- **Step 3.** Substitute each endpoint into the series and test the resulting number series (alternating test, p-series, nth-term test…).
- **Step 4.** State the interval with correct brackets: [ ] included, ( ) excluded.



Three intervals of convergence. Filled dot: endpoint converges; open dot: endpoint diverges.

WORKED EXAMPLE

(a) $\Sigma x^n/n$. Ratio = $|x| \cdot n/(n+1) \to |x|$. R = 1. At x = 1: $\Sigma 1/n$ diverges. At x = −1: $\Sigma(-1)^n/n$ converges (alternating). Interval **[−1, 1)**.

(b) $\Sigma x^n/n^2$. R = 1. Both endpoints give p-series with p = 2 (absolutely convergent). Interval **[−1, 1]**.

(c) $\Sigma x^n/n!$. Ratio = $|x|/(n+1) \to 0$ for every x. **R = ∞**, interval $(-\infty, \infty)$.

(d) $\Sigma n!\, x^n$. Ratio = $(n+1)|x| \to \infty$ unless x = 0. **R = 0**: converges only at x = 0.

(e) $\Sigma (x-2)^n/(n \cdot 3^n)$. Ratio $\to |x-2|/3 < 1 \Rightarrow$ R = 3, centre 2, open interval (−1, 5). At x = 5: $\Sigma 1/n$ diverges; at x = −1: $\Sigma(-1)^n/n$ converges. Interval **[−1, 5)**.

(f) $\Sigma 2^n x^n/\sqrt{n}$. Ratio $\to 2|x| < 1 \Rightarrow$ R = 1/2. At x = 1/2: $\Sigma 1/\sqrt{n}$ diverges; at x = −1/2: converges (alternating). Interval **[−1/2, 1/2)**.

(g) $\Sigma x^{2n}/4^n$ (only even powers). Ratio on the whole term = $x^2/4 < 1 \Rightarrow |x| < 2$. At x = ±2 the terms equal 1 and do not → 0, so both diverge. Interval **(−2, 2)**, R = 2.

WATCH OUT: EXAM TRAP

With missing powers (only x^{2n} or x^{2n+1}), apply the ratio test to the whole term, not just the coefficients. Then x^2 appears and you must take a square root to get R.

Never skip the endpoints. The ratio test is always inconclusive there (L = 1), and the interval is incomplete without them.

5.2 Working with power series

THE IDEA IN PLAIN ENGLISH

Inside the open interval (a − R, a + R) a power series behaves like a polynomial. You can **add, multiply, differentiate and integrate term by term**, and the **radius stays the same** (endpoint behaviour may change). This lets us build new series from old ones.

WORKED EXAMPLE

Start from the geometric series: $1/(1-x) = 1 + x + x^2 + x^3 + \ldots = \Sigma x^n$, $|x| < 1$.

Substitute x → −x: $1/(1+x) = 1 - x + x^2 - x^3 + \ldots = \Sigma(-1)^n x^n$.

Integrate term by term: $\ln(1+x) = x - x^2/2 + x^3/3 - \ldots = \mathbf{\Sigma(-1)^{n+1}\, x^n/n}$, $-1 < x \leq 1$. (At x = 1: $1 - 1/2 + 1/3 - \ldots = \ln 2$.)

Substitute x → −x²: $1/(1+x^2) = \Sigma(-1)^n x^{2n}$. **Integrate:** $\arctan x = \mathbf{\Sigma(-1)^n\, x^{2n+1}/(2n+1)}$, $|x| \leq 1$. (At x = 1: $\pi/4 = 1 - 1/3 + 1/5 - \ldots$)

Differentiate the geometric series: $1/(1-x)^2 = 1 + 2x + 3x^2 + \ldots = \Sigma(n+1)x^n$.

A series for 1/(2 − x): $= (1/2)\cdot 1/(1 - x/2) = \Sigma\, x^n/2^{n+1}$, valid for $|x| < 2$.

JOT THIS DOWN

Power series: $\Sigma c_n(x-a)^n$. Always R = 0, R = ∞, or a finite R.

Find R: ratio test on the whole term → $|x - a| < R$. **Then test both endpoints separately.**

Interval notation: [ ] converges at the endpoint, ( ) does not.

Term-by-term differentiate/integrate inside (a − R, a + R); R unchanged.

Build new series: $1/(1-x) = \Sigma x^n \to \ln(1+x) = \Sigma(-1)^{n+1}x^n/n \to \arctan x = \Sigma(-1)^n x^{2n+1}/(2n+1)$.

PART 6 | Taylor and Maclaurin Series; Binomial Theorem

6.1 The idea

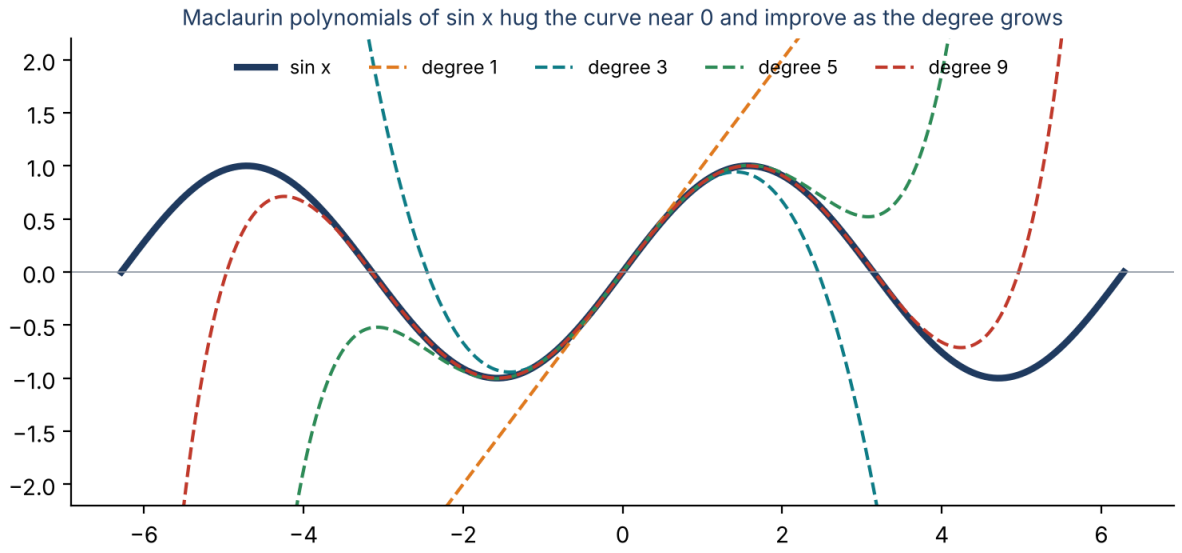
THE IDEA IN PLAIN ENGLISH

Polynomials are easy: you can evaluate them with just +, −, ×. Functions like sin x, e^x and ln x are not. **A Taylor series rewrites a complicated function as an infinite polynomial** built to match the function perfectly at one point a: the same **value**, the same **slope**, the same **curvature**, and so on for every derivative at a.

Why the n! appears: if $f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \ldots$, differentiating n times and setting x = a kills every term except $c_n \cdot n!$. So $f^{(n)}(a) = n! \cdot c_n$, i.e. $c_n = f^{(n)}(a)/n!$.

$$\mathbf{f(x) = \Sigma_{n=0}^{\infty}\, [f^{(n)}(a)/n!]\,(x-a)^n = f(a) + f'(a)(x-a) + f''(a)(x-a)^2/2! + f'''(a)(x-a)^3/3! + \ldots}$$

- **Maclaurin series** = Taylor series with **a = 0**: $f(x) = \Sigma\, f^{(n)}(0)\, x^n/n!$.
- **Taylor polynomial of degree n**, $P_n(x)$: the first n + 1 terms. It is an approximation; the leftover is the **remainder** $R_n(x) = f(x) - P_n(x)$.
- **Lagrange form of the remainder:** $R_n(x) = f^{(n+1)}(c)/(n+1)! \cdot (x - a)^{n+1}$ for some c between a and x. The series equals f(x) exactly when $R_n \to 0$.



Higher-degree Maclaurin polynomials track sin x over a wider and wider range.

6.2 Deriving a Maclaurin series from the formula

WORKED EXAMPLE

f(x) = e^x: every derivative is e^x, so $f^{(n)}(0) = 1$. Series: **1 + x + x^2/2! + x^3/3! + …**

f(x) = sin x: derivatives at 0 cycle through **0, 1, 0, −1, 0, 1, …** (sin, cos, −sin, −cos). Only odd powers survive, with alternating signs: **x − x^3/3! + x^5/5! − …**

f(x) = cos x: derivatives at 0 cycle **1, 0, −1, 0, …**: **1 − x^2/2! + x^4/4! − …** (only even powers).

6.3 Standard Maclaurin series (memorise these)

Function	Series	Valid for
e^x	$\Sigma\, x^n/n! = 1 + x + x^2/2! + x^3/3! + \ldots$	all x
sin x	$\Sigma(-1)^n\, x^{2n+1}/(2n+1)! = x - x^3/3! + x^5/5! - \ldots$	all x
cos x	$\Sigma(-1)^n\, x^{2n}/(2n)! = 1 - x^2/2! + x^4/4! - \ldots$	all x
sinh x	$\Sigma\, x^{2n+1}/(2n+1)! = x + x^3/3! + x^5/5! + \ldots$	all x
cosh x	$\Sigma\, x^{2n}/(2n)! = 1 + x^2/2! + x^4/4! + \ldots$	all x
$1/(1 - x)$	$\Sigma\, x^n = 1 + x + x^2 + x^3 + \ldots$	$\lvert x\rvert < 1$
$1/(1 + x)$	$\Sigma(-1)^n x^n = 1 - x + x^2 - \ldots$	$\lvert x\rvert < 1$
ln(1 + x)	$\Sigma(-1)^{n+1}\, x^n/n = x - x^2/2 + x^3/3 - \ldots$	$-1 < x \le 1$
arctan x	$\Sigma(-1)^n\, x^{2n+1}/(2n+1) = x - x^3/3 + x^5/5 - \ldots$	$\lvert x\rvert \le 1$
$(1 + x)^\alpha$	$\Sigma\, C(\alpha, n)\, x^n = 1 + \alpha x + \alpha(\alpha-1)x^2/2! + \ldots$	$\lvert x\rvert < 1$

THE IDEA IN PLAIN ENGLISH

Memory help. The series of e^x, sin x and cos x are linked: sin x and cos x are the odd and even parts of e^{ix}. Sin is odd (only odd powers), cos is even (only even powers), and the signs alternate because the derivatives cycle with period 4. To get a new series, **substitute** into a known one (e.g. replace x by 2x, $-x^2$, etc.) instead of differentiating again.

6.4 Worked examples

WORKED EXAMPLE

(a) e^{2x}: substitute 2x for x: $\Sigma(2x)^n/n! = \mathbf{1 + 2x + 2x^2 + (4/3)x^3 + \ldots}$ (all x).

(b) e^{3x} up to x^3: $1 + 3x + 9x^2/2 + 27x^3/6 = \mathbf{1 + 3x + (9/2)x^2 + (9/2)x^3}$.

(c) x cos x: multiply the cos series by x: $\mathbf{x - x^3/2 + x^5/24 - \ldots}$

(d) $\cos(x^2)$: replace x by x^2: $\mathbf{1 - x^4/2! + x^8/4! - \ldots}$

(e) e^{-x^2}: $1 - x^2 + x^4/2 - x^6/6 + \ldots$

(f) $\sin^2 x$ using $\sin^2 x = (1 - \cos 2x)/2$: $\cos 2x = 1 - 2x^2 + (2/3)x^4 - (4/45)x^6 + \ldots$, so $\sin^2 x = \mathbf{x^2 - x^4/3 + 2x^6/45 - \ldots}$

(g) Taylor series of $\sqrt{x}$ about a = 4 (degree 2). $f = \sqrt{x}$, $f' = 1/(2\sqrt{x})$, $f'' = -1/(4x^{3/2})$. At 4: $f = 2$, $f' = 1/4$, $f'' = -1/32$. So $P_2(x) = 2 + (x - 4)/4 - (x - 4)^2/64$. Estimate $\sqrt{4.1}$: $2 + 0.025 - 0.00015625 =$ **2.02484** (calculator: 2.024846).

(h) 1/x about a = 1: write $x = 1 + (x - 1)$: $1/x = 1/(1 + (x - 1)) = \mathbf{\Sigma(-1)^n(x - 1)^n}$ for $|x - 1| < 1$.

(i) ln x about a = 1: integrate (h): $\mathbf{\Sigma(-1)^{n+1}(x - 1)^n/n}$.

6.5 Applications

Limits

WORKED EXAMPLE

lim $(\sin x - x)/x^3$: $\sin x - x = -x^3/6 + x^5/120 - \ldots \Rightarrow$ divide by $x^3 \to \mathbf{-1/6}$.

lim $(1 - \cos x)/x^2$: $1 - \cos x = x^2/2 - x^4/24 + \ldots \Rightarrow \mathbf{1/2}$.

lim $(e^x - 1 - x)/x^2$: numerator $= x^2/2 + x^3/6 + \ldots \Rightarrow \mathbf{1/2}$.

lim $(x - \ln(1 + x))/x^2$: $\ln(1 + x) = x - x^2/2 + x^3/3 \Rightarrow$ numerator $= x^2/2 - x^3/3 + \ldots \Rightarrow \mathbf{1/2}$.

lim $(\cos x - 1 + x^2/2)/x^4$: $\cos x = 1 - x^2/2 + x^4/24 - \ldots \Rightarrow \mathbf{1/24}$.

Approximating numbers and integrals, with error control

WORKED EXAMPLE

sin 0.1: $0.1 - (0.1)^3/6 = 0.099833\ldots$ The series alternates, so the error is less than the next term $(0.1)^5/120 \approx 8 \times 10^{-8}$.

$e^{0.1}$: $1 + 0.1 + 0.005 + 0.000167 = 1.105167$. Lagrange error $\le e^c(0.1)^4/4! \le 1.2 \times 10^{-4}/24 = 5 \times 10^{-6}$. (True value 1.105171.)

How many terms of the series for e give error < 10^{-5}? Lagrange: $R_n \le e/(n+1)! < 3/(n+1)!$. Need $3/(n+1)! < 10^{-5}$: n = 8 gives $3/9! = 8.3 \times 10^{-6}$ ✓. So $1 + 1 + 1/2 + \ldots + 1/8! =$ **2.71828**.

$\int_0^1 e^{-x^2}\,dx$ (no elementary antiderivative): integrate $1 - x^2 + x^4/2 - x^6/6 + \ldots$ term by term: $1 - 1/3 + 1/10 - 1/42 + 1/216 - 1/1320 \approx$ **0.7468**.

$\int_0^{0.5} \cos(x^2)\,dx$: $\int(1 - x^4/2 + x^8/24) = 0.5 - (0.5)^5/10 + (0.5)^9/216 = 0.5 - 0.003125 + 0.000009 \approx$ **0.4969**.

6.6 The binomial theorem

THE IDEA IN PLAIN ENGLISH

For a positive whole number n: expanding $(a + b)^n$ gives n + 1 terms. The **binomial coefficient** $C(n, k) = n!/[k!(n - k)!]$ counts how many ways to pick k of the n brackets to supply b.

$(a + b)^n = \Sigma_{k=0}^{n} C(n, k)\, a^{n-k}\, b^k$. The **general term** is **$T_{k+1} = C(n, k)\, a^{n-k}\, b^k$**.

- **Pascal's triangle** gives the coefficients: row n = 1, n, …, n, 1 (each entry is the sum of the two above it). Row 4: 1 4 6 4 1. Row 5: 1 5 10 10 5 1.
- Number of terms = n + 1. Sum of all coefficients of $(1 + x)^n = 2^n$. Coefficients are symmetric: $C(n, k) = C(n, n - k)$.
- **Middle term:** n even → one middle term, $T_{n/2+1}$; n odd → two middle terms.

WORKED EXAMPLE

Expand $(1 + 2x)^4$: $C(4, k)2^k x^k$: $1 + 8x + 24x^2 + 32x^3 + 16x^4$.

Coefficient of x^3 in $(2 + x)^6$: $C(6, 3)\cdot 2^3 = 20 \times 8 =$ **160**.

Coefficient of x^4 in $(3 - 2x)^7$: $C(7, 4)\cdot 3^3\cdot(-2)^4 = 35 \times 27 \times 16 =$ **15120**.

Term independent of x in $(x^2 + 1/x)^9$: $T_{k+1} = C(9, k)(x^2)^{9-k}(1/x)^k = C(9, k)x^{18-3k}$. Set $18 - 3k = 0 \Rightarrow k = 6$. Term = $C(9, 6) =$ **84**.

The binomial series (any real power α)

$$(1 + x)^{\alpha} = 1 + \alpha x + \alpha(\alpha - 1)x^2/2! + \alpha(\alpha - 1)(\alpha - 2)x^3/3! + \ldots \quad |x| < 1$$

THE IDEA IN PLAIN ENGLISH

When α is a positive integer the series stops (it is just the binomial theorem). When α is a fraction or negative it **never stops** and converges only for $|x| < 1$. This is how you expand square roots and reciprocals.

WORKED EXAMPLE

$(1 + x)^{1/2} = 1 + x/2 - x^2/8 + x^3/16 - 5x^4/128 + \ldots$ Check: $C(1/2, 2) = (1/2)(-1/2)/2 = -1/8$.

$(1 - x)^{-1/2} = 1 + x/2 + 3x^2/8 + 5x^3/16 + \ldots$

Estimate $\sqrt{1.04}$: $(1 + 0.04)^{1/2} \approx 1 + 0.02 - 0.0002 =$ **1.0198** (calculator 1.019804).

Estimate $\sqrt{1.02} \approx 1 + 0.01 - 0.00005 =$ 1.00995.

JOT THIS DOWN

Taylor: $f(x) = \Sigma\, f^{(n)}(a)(x - a)^n/n!$. **Maclaurin:** a = 0.

Remainder (Lagrange): $R_n = f^{(n+1)}(c)(x - a)^{n+1}/(n+1)!$. Use it for error bounds.

Must memorise: e^x, sin x, cos x, $1/(1-x)$, $\ln(1+x)$, arctan x, $(1+x)^{\alpha}$ (table in 6.3).

Shortcut: substitute into known series (e^{-x^2}, cos x^2, …); multiply/differentiate/integrate term by term.

Limits: expand numerator and denominator, cancel the lowest power.

Binomial: $(a+b)^n = \Sigma C(n,k)a^{n-k}b^k$; $T_{k+1} = C(n,k)a^{n-k}b^k$. Series: $(1+x)^{\alpha} = \Sigma C(\alpha,n)x^n$, $|x| < 1$.

PART 7 | Common Mistakes and Strategy

Mistake	Fix
"$a_n \to 0$, so Σa_n converges"	False (harmonic series). It only shows the nth-term test is inconclusive.
Using the ratio test on $\Sigma 1/n^p$ or rational terms	$L = 1$ tells you nothing. Use limit comparison with a p-series.
Wrong comparison direction	Smaller than convergent ⇒ converges. Bigger than divergent ⇒ diverges. Nothing else works.
Geometric sum with wrong first term	Always write (first term)/(1 − r); check where n starts.
Forgetting endpoints of a power series	Substitute each endpoint and test the number series.
Calling a conditionally convergent series "absolutely convergent"	Test $\Sigma \lvert a_n \rvert$ separately.
Solving L = f(L) before proving convergence	Show monotone and bounded first.
$\Sigma(a_n b_n) = \Sigma a \Sigma b$	Wrong. Only sums and constant multiples split.
Wrong Σ index count	Number of terms = (top − bottom) + 1.
Missing the (-1) in alternating signs	Check the first terms against the series.
Dropping n! in Taylor coefficients	$c_n = f^{(n)}(a)/\mathbf{n!}$.
Quoting the Maclaurin series outside its interval	Check validity: ln(1+x) needs $-1 < x \leq 1$.

PART 8 | Test Yourself

Try each without looking. Answers on the next page. Questions 1 to 6 are sequences, 7 to 14 progressions and finite sums, 15 to 20 series and tests, 21 to 24 power series, 25 to 32 Taylor, Maclaurin and binomial, 33 to 36 theory.

1. Find the general term: (a) 3, 6, 9, 12, … (b) 2, −4, 8, −16, … (c) 2/3, 3/5, 4/7, …

2. Find the limit, or show it diverges: (a) $(5n^2 - 3)/(2n^2 + n)$ (b) $\sqrt{(n+1)} - \sqrt{n}$ (c) $(-1)^n n/(n+1)$ (d) $(1 + 3/n)^n$ (e) $n^2/2^n$ (f) $(2^n + 5^n)^{1/n}$

3. Prove from the ε–N definition that $(3n + 1)/n \to 3$.

4. Show that $a_n = n/(n^2 + 1)$ is decreasing and bounded. What is its limit?

5. $a_1 = \sqrt{6}$, $a_{n+1} = \sqrt{(6 + a_n)}$. Show it converges and find the limit.

6. State the Monotone Convergence Theorem. Why does $(-1)^n$ not contradict it?

7. In an AP, $T_4 = 11$ and $T_{10} = 29$. Find a, d and S_{20}.

8. Find the sum of the first 8 terms of the GP 5, 10, 20, …

9. Express 0.151515… as a fraction.

10. Three numbers in AP have sum 21 and product 315. Find them.

11. Evaluate $\Sigma_{k=1}^{20} (3k - 2)$.

12. Find $\Sigma_{k=1}^{n} 1/[k(k+2)]$ and its limit as $n \to \infty$.

13.	Prove by induction that $1 + 3 + 5 + \ldots + (2n - 1) = n^2$.
14.	Find the AM, GM and HM of 2 and 8 and verify $GM^2 = AM \times HM$. Insert 3 arithmetic means between 4 and 20.
15.	Sum: (a) $\Sigma_{n=0}^{\infty} 5(-1/2)^n$ (b) $\Sigma_{n=1}^{\infty} (1/\sqrt{n} - 1/\sqrt{(n+1)})$.
16.	Show $\Sigma (2n^2 + 1)/(3n^2 + n)$ diverges.
17.	Decide convergence and name the test: (a) $\Sigma (n + 2)/(n^3 + 1)$ (b) $\Sigma n!/3^n$ (c) $\Sigma (n/(2n + 1))^n$ (d) $\Sigma 1/(n \ln^2 n)$, $n \geq 2$ (e) $\Sigma 1/(\sqrt{n} + 1)$
18.	Classify as absolutely convergent, conditionally convergent or divergent: (a) $\Sigma(-1)^n/(2n + 1)$ (b) $\Sigma(-1)^n/n^3$ (c) $\Sigma(-1)^n \ln n / n$
19.	$\Sigma(-1)^{n+1}/n^2$ is approximated by its first 4 terms. Bound the error.
20.	Does $\Sigma(-1)^n n/(n^2 + 1)$ converge absolutely, conditionally, or diverge?
21.	True or false (explain): if $a_n \to 0$ then Σa_n converges.
22.	Find the radius and interval of convergence: (a) $\Sigma (x + 1)^n/(n \cdot 2^n)$ (b) $\Sigma n! x^n$ (c) $\Sigma (3x)^n/\sqrt{n}$ (d) $\Sigma x^{2n}/4^n$ (e) $\Sigma x^n/n!$
23.	Find a power series for $1/(2 - x)$ and its interval of convergence.
24.	Obtain a power series for $\ln(1 + x^2)$ from the series for $\ln(1 + x)$.
25.	Starting from $1/(1 + x^2)$, derive the series for $\arctan x$ and use it to write $\pi/4$ as a series.
26.	Write the first four non-zero terms of the Maclaurin series of e^{3x}.
27.	Maclaurin series of $\cos(x^2)$ (first three terms).
28.	Taylor series of $f(x) = 1/x$ about $a = 1$, with interval of validity.
29.	Evaluate $\lim (\cos x - 1 + x^2/2)/x^4$ using series.
30.	Approximate $\int_0^{0.5} \cos(x^2)\, dx$ using the first two non-zero terms of the series.
31.	Find the coefficient of x^4 in $(3 - 2x)^7$ and the term independent of x in $(2x - 1/x^2)^9$.
32.	Use the binomial series to estimate $\sqrt{1.04}$ to 4 decimal places.
33.	Taylor polynomial of degree 2 for $\sqrt{x}$ about $a = 4$, and use it to estimate $\sqrt{4.1}$.
34.	Why is the radius of convergence of the Maclaurin series of $1/(1 + x^2)$ equal to 1, even though the function is smooth for all real x?
35.	Define (a) absolutely convergent series (b) conditionally convergent series (c) radius of convergence.
36.	Distinguish between a sequence and a series with an example of each.
37.	Why must $\Sigma\lvert a_n\rvert$ be tested separately from Σa_n? What do we conclude if $\Sigma\lvert a_n\rvert$ converges?

Answers

1. (a) **3n** (b) $\mathbf{(-1)^{n+1}\,2^n}$ (c) $\mathbf{(n+1)/(2n+1)}$.

2. (a) **5/2** (divide by n^2). (b) $= 1/(\sqrt{(n+1)} + \sqrt{n}) \to \mathbf{0}$. (c) **diverges**: even n gives $\to 1$, odd n gives $\to -1$. (d) $\mathbf{e^3}$. (e) **0** (exponential beats polynomial). (f) $5 \le (2^n + 5^n)^{1/n} \le 5\cdot 2^{1/n} \to 5$, so **5**.

3. $|(3n+1)/n - 3| = 1/n < \varepsilon$ whenever $n > 1/\varepsilon$. Take $N = 1/\varepsilon$. ∎

4. $f(x) = x/(x^2+1)$, $f'(x) = (1-x^2)/(x^2+1)^2 \le 0$ for $x \ge 1$, so decreasing. $0 < a_n \le 1/2$, so bounded. Limit = **0** ($a_n < 1/n$).

5. Bounded above by 3: $\sqrt{6} < 3$, and if $a_n < 3$ then $a_{n+1} < \sqrt{9} = 3$. Increasing: $a_{n+1}^2 - a_n^2 = (3 - a_n)(2 + a_n) > 0$. Converges to $L = \sqrt{(6+L)} \Rightarrow L^2 - L - 6 = 0 \Rightarrow L = \mathbf{3}$ (reject −2).

6. Every bounded monotonic sequence converges. $(-1)^n$ is bounded but **not monotonic**.

7. $d = (29-11)/6 = \mathbf{3}$; $a = 11 - 9 = \mathbf{2}$; $S_{20} = (20/2)[4 + 19(3)] = \mathbf{610}$.

8. $a = 5$, $r = 2$: $S_8 = 5(2^8 - 1) = 5 \times 255 = \mathbf{1275}$.

9. $0.15 + 0.0015 + \ldots$: $a = 0.15$, $r = 0.01 \Rightarrow 0.15/0.99 = 15/99 = \mathbf{5/33}$.

10. Let them be $a-d$, a, $a+d$: $3a = 21 \Rightarrow a = 7$. $(7-d)(7)(7+d) = 315 \Rightarrow 49 - d^2 = 45 \Rightarrow d = \pm 2$. Numbers: **5, 7, 9**.

11. $\Sigma(3k-2) = 3(210) - 40 = \mathbf{590}$. (Check: AP 1, 4, …, 58: $(20/2)(1+58) = 590$.)

12. $1/[k(k+2)] = \tfrac{1}{2}(1/k - 1/(k+2))$. Sum $= \tfrac{1}{2}[1 + 1/2 - 1/(n+1) - 1/(n+2)] \to \mathbf{3/4}$.

13. Base $n = 1$: $1 = 1^2$ ✓. Assume $1 + 3 + \ldots + (2k-1) = k^2$. Add $(2k+1)$: $k^2 + 2k + 1 = (k+1)^2$ ✓. ∎

14. AM = 5, GM = 4, HM $= 2\cdot 16/10 = 3.2$. $GM^2 = 16 = 5 \times 3.2$ ✓. Three AMs between 4 and 20: $d = 16/4 = 4$ ⇒ **8, 12, 16**.

15. (a) $5/(1 + 1/2) = \mathbf{10/3}$. (b) telescopes: $S_n = 1 - 1/\sqrt{(n+1)} \to \mathbf{1}$.

16. $a_n \to 2/3 \ne 0$, so by the **nth-term test** it diverges.

17. (a) limit comparison with $1/n^2$ (limit 1): **converges**. (b) ratio $= (n+1)/3 \to \infty$: **diverges**. (c) root $= n/(2n+1) \to 1/2$: **converges**. (d) integral: $\int dx/(x \ln^2 x) = -1/\ln x$ finite: **converges**. (e) limit comparison with $1/\sqrt{n}$ (limit 1): **diverges**.

18. (a) **Conditional** ($\Sigma 1/(2n+1)$ diverges by comparison with $1/n$; AST holds). (b) **Absolute** ($p = 3$). (c) **Conditional** ($\ln n/n \ge 1/n$ for $n \ge 3$ so $\Sigma|a_n|$ diverges; AST holds for $n \ge 3$).

19. Error ≤ first omitted term $= 1/5^2 = \mathbf{0.04}$.

20. $|a_n| = n/(n^2+1) \sim 1/n$, so $\Sigma|a_n|$ diverges (limit comparison). Terms decrease ($n \ge 1$) and $\to 0$, so the AST gives convergence: **conditionally convergent**.

21. **False.** The harmonic series: $1/n \to 0$ but $\Sigma 1/n$ diverges.

22. (a) ratio $\to |x+1|/2$, $R = 2$, centre -1. $x = 1$: $\Sigma 1/n$ diverges; $x = -3$: converges. **$[-3, 1)$**. (b) **$R = 0$**, only $x = 0$. (c) $R = 1/3$; $x = 1/3$ diverges ($\Sigma 1/\sqrt{n}$), $x = -1/3$ converges. **$[-1/3, 1/3)$**. (d) $x^2/4 < 1$: $R = 2$; at ± 2 terms = 1, diverge. **$(-2, 2)$**. (e) **$R = \infty$**, $(-\infty, \infty)$.

23. $1/(2-x) = \tfrac{1}{2}\cdot 1/(1 - x/2) = \mathbf{\Sigma\, x^n/2^{n+1}}$, valid for $|x/2| < 1$, i.e. **$|x| < 2$** (interval $(-2, 2)$).

24. Replace x by x^2: $\ln(1 + x^2) = \mathbf{\Sigma(-1)^{n+1}\, x^{2n}/n = x^2 - x^4/2 + x^6/3 - \ldots}$, $|x| \le 1$.

25. $1/(1 + x^2) = \Sigma(-1)^n x^{2n}$. Integrate: $\arctan x = \mathbf{\Sigma(-1)^n\, x^{2n+1}/(2n+1)}$. At $x = 1$: **$\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \ldots$**

26. **$1 + 3x + (9/2)x^2 + (9/2)x^3 + \ldots$** (next term: $27x^4/8$).

27. **$1 - x^4/2 + x^8/24 - \ldots$**

28.	$1/x = 1/(1 + (x - 1)) = \mathbf{\Sigma(-1)^n(x-1)^n}$, valid for $\mathbf{|x - 1| < 1}$, i.e. $0 < x < 2$.
29.	$\cos x = 1 - x^2/2 + x^4/24 - \ldots$, so numerator $= x^4/24 - x^6/720 + \ldots$; divided by $x^4 \to$ **1/24**.
30.	$\int_0^{0.5} (1 - x^4/2)\,dx = 0.5 - (0.5)^5/10 = 0.5 - 0.003125 =$ **0.4969** (rounded).
31.	x^4: $C(7, 4)\cdot 3^3\cdot(-2)^4 = 35 \times 27 \times 16 =$ **15120**. Independent term: $T_{k+1} = C(9, k)(2x)^{9-k}(-1/x^2)^k \propto x^{9-3k}$; $k = 3$: $C(9, 3)\cdot 2^6\cdot(-1)^3 = 84 \times 64 \times (-1) =$ **−5376**.
32.	$(1 + 0.04)^{1/2} = 1 + 0.02 - (0.04)^2/8 + (0.04)^3/16 = 1 + 0.02 - 0.0002 + 0.000004 =$ **1.0198**.
33.	$P_2(x) = 2 + (x - 4)/4 - (x - 4)^2/64$. $\sqrt{4.1} \approx 2 + 0.025 - 0.00015625 =$ **2.02484**.
34.	The function has singularities at $x = \pm i$ in the complex plane (where $1 + x^2 = 0$), and those lie at distance 1 from the centre 0. The radius of convergence is the distance to the nearest singularity, even a complex one.
35.	(a) $\Sigma\|a_n\|$ converges. (b) Σa_n converges but $\Sigma\|a_n\|$ diverges. (c) R such that a power series converges for $\|x - a\| < R$ and diverges for $\|x - a\| > R$.
36.	A **sequence** is an ordered list (1, 1/2, 1/3, …); a **series** is the sum of its terms (1 + 1/2 + 1/3 + …), analysed through its sequence of partial sums.
37.	Convergence of Σa_n and $\Sigma\|a_n\|$ are different questions (conditional convergence). If $\Sigma\|a_n\|$ converges then Σa_n converges too (absolute convergence ⇒ convergence).

PART 9 | One-Page Formula Sheet

Topic	Formula / rule
AP	$T_n = a + (n-1)d$; $S_n = (n/2)[2a + (n-1)d]$
GP	$T_n = ar^{n-1}$; $S_n = a(r^n-1)/(r-1)$; $S_\infty = a/(1-r)$, $\lvert r\rvert < 1$
Means	AM = (a+b)/2; GM = $\sqrt{(ab)}$; HM = 2ab/(a+b); AM ≥ GM ≥ HM
Standard sums	$\Sigma k = n(n+1)/2$; $\Sigma k^2 = n(n+1)(2n+1)/6$; $\Sigma k^3 = [n(n+1)/2]^2$
Standard limits	$r^n \to 0$ ($\lvert r\rvert<1$); $n^{1/n} \to 1$; $(1+x/n)^n \to e^x$; $x^n/n! \to 0$; $\ln n \ll n^p \ll a^n \ll n! \ll n^n$
Convergent sequence	Monotone + bounded ⇒ convergent
Series	Σa_n converges ⇔ S_n converges; $a_n \not\to 0$ ⇒ diverges
Geometric / p-series	Σar^n = first/(1−r), $\lvert r\rvert<1$; $\Sigma 1/n^p$ converges ⇔ p > 1
Tests	Integral; Comparison; Limit comparison; Ratio ($\lim\lvert a_{n+1}/a_n\rvert$); Root ($\lim\lvert a_n\rvert^{1/n}$); Alternating ($b_n\downarrow$, $\to 0$, error ≤ b_{n+1})
Absolute / conditional	$\Sigma\lvert a_n\rvert$ converges ⇒ Σa_n converges. Conditional: Σa_n converges, $\Sigma\lvert a_n\rvert$ diverges
Power series	$\Sigma c_n(x-a)^n$; ratio test ⇒ R; test endpoints separately
Taylor	$f(x) = \Sigma f^{(n)}(a)(x-a)^n/n!$; $R_n = f^{(n+1)}(c)(x-a)^{n+1}/(n+1)!$
Maclaurin	$e^x = \Sigma x^n/n!$; $\sin x = \Sigma(-1)^n x^{2n+1}/(2n+1)!$; $\cos x = \Sigma(-1)^n x^{2n}/(2n)!$
More Maclaurin	$1/(1-x) = \Sigma x^n$; $\ln(1+x) = \Sigma(-1)^{n+1}x^n/n$; $\arctan x = \Sigma(-1)^n x^{2n+1}/(2n+1)$
Binomial	$(a+b)^n = \Sigma C(n,k)a^{n-k}b^k$; $(1+x)^\alpha = \Sigma C(\alpha,n)x^n$, $\lvert x\rvert<1$

SIX MISTAKES TO AVOID

1. $a_n \to 0$ does NOT mean the series converges.
2. Ratio/root test with L = 1 is inconclusive.
3. Check the comparison inequality direction.
4. Geometric sum: (first term)/(1 − r); check the start index.
5. Test power-series endpoints separately.
6. Prove convergence before solving L = f(L).